# Viztalent AI

How questions are created, how candidates are tested and interviewed, and where AI is - and is not - used

Based on a direct analysis of the production code (backend-classic + Viztalent-ai-frontend, main branch, October 2026). Every rule, number and threshold below is taken from the source code, not from earlier design documents.

## At a glance

| What | How many | AI needed? |
|---|---|---|
| Hiring stages | 6 - role profile, resume screening, online assessment, AI interview, grading, expert review | Partly |
| Question types in assessments | 7 - MCQ, MSQ, Code-Write, Code-Debug, Code-Output, Scenario, Logic | - |
| Assessment levels | 3 - L1, L2, L3 (each a 100-point test) | - |
| Ways to build an assessment | 4 - AI, Mix, Custom (own question bank), Personal (per candidate) | Custom: no; others: yes |
| Interview channels | 3 - Browser (recruiter-run), Self-service link, Phone call (Twilio) | Yes, always |
| Final decisions | 3 - PROCEED, HR_REVIEW, NOT_RECOMMENDED | No (rules) |

# 1. The hiring pipeline

A job moves through six stages. AI (Google Gemini) helps in some of them; others are pure rules that work with no AI at all.

| # | Stage | What happens | AI? |
|---|---|---|---|
| 1 | Job + role profile | The recruiter uploads a JD (or pastes skills). A role profile is built: role family, technical or not, seniority, languages, and a weight for every skill. | AI, with a rule-based fallback |
| 2 | Resume screening | Uploaded resumes are scored against the JD skills and experience and ranked. | No AI |
| 3 | Online assessment | Candidates take a timed, proctored test (L1/L2/L3) from a single-use link. | Questions: AI or bank |
| 4 | AI interview | Spoken Q&A; in the browser, via a self-service link, or by phone. | Yes |
| 5 | Grading + scoring | Objective answers are checked by rules/code; written and spoken answers by an AI rubric. Scores become Theory / Hands-on and a decision. | Partly |
| 6 | Expert (SME) review | A technical reviewer can approve question sets and sign off results. | No AI (human) |

## 1.1 Role profile (stage 1)

Right after a JD is uploaded, the backend asks Gemini to read it and return a structured profile: role family (backend, frontend, data, non-technical ...), whether the role is technical, seniority, programming languages/frameworks, and a weight per skill. Gemini may only keep skills that were actually extracted from the JD.

**Without AI:** a rule-based profile is built from the JD's extracted skills and the experience range (seniority comes from the minimum years). Everything downstream still works.

## 1.2 Resume screening (stage 2) - no AI

Screening is deterministic, so the same resumes always get the same score:

- Required skills matched: up to **85 points** (share of the JD's must-have skills found in the resume).
- Nice-to-have skills matched: up to **15 points**.
- Experience: **+5** if the resume's years meet the JD minimum, **-5** if not.
- Candidates scoring **50 or more** are counted as shortlisted. Each row shows which skills matched and which are missing.

The recruiter chat assistant on the Screening page uses Gemini to understand requests such as "screen these resumes", but the ranking itself is this rule-based score.

# 2. Assessment questions

## 2.1 The seven question types

Every question belongs to one of two dimensions - **Theory** (knowing) or **Hands-on** (doing) - which later become the two headline scores.

| Type | What the candidate does | Dimension | How it is graded |
|---|---|---|---|
| MCQ | Pick one correct option | Theory | Rules (exact match) - no AI |
| MSQ | Pick all correct options | Theory | Rules - no AI |
| CODE_WRITE | Write a program in a supported language (Java, Python, JavaScript, C++) | Hands-on | 70% hidden tests run in a sandbox (no AI) + 30% AI rubric on code quality |
| CODE_DEBUG | Fix buggy code so it passes the tests | Hands-on | Hidden tests in the sandbox - no AI |
| CODE_OUTPUT | Predict what a snippet prints | Hands-on | Rules - no AI |
| SCENARIO | Written answer: real situations, design, debugging reasoning, DevOps, Agile | Hands-on | AI rubric against key points (100%) |
| LOGIC | Logical / aptitude question (used for junior roles) | Hands-on | Rules - no AI |

Question banks uploaded by HR (JSON, CSV or XLSX) may use everyday labels - "Subjective", "Code & Bug", "Logical / Output", "Multiple choice" - which are mapped onto these seven types automatically.

## 2.2 Levels and blueprints

Each level is a **100-point blueprint**: a fixed list of question slots, each with a type, points, difficulty and a target skill. Skills are assigned with a weighted round-robin, so heavily weighted skills get proportionally more questions; coding slots only target programming skills.

| Level | Technical roles | Non-technical roles |
|---|---|---|
| L1 (~45 min, easy) | 4 MCQ + 2 MSQ (5 pts each = 30) - Code-Write easy 15 + medium 20 - Code-Debug 15 - 2 Code-Output x5 - Scenario 10 (Logic for juniors) | 6 MCQ + 2 MSQ (40) - 3 Scenarios x20 (60) |
| L2 (medium) | 3 MCQ + 1 MSQ (20) - Code-Write medium 18 + hard 22 - Code-Debug 15 - Code-Output 10 - design Scenario 15 | 4 MCQ + 2 MSQ (30) - Scenarios 20 + 25 + 25 |
| L3 (hard) | 2 MCQ + 1 MSQ (15) - 2 Code-Write hard x20 - Code-Debug 15 - Code-Output 10 - design Scenario 20 | 3 MCQ + 1 MSQ (20) - 4 Scenarios x20 |

Time per question is estimated (e.g. MCQ 1 min, hard Code-Write 15 min, design Scenario 10 min) and summed into the test's duration.

## 2.3 Four ways an assessment is built

| Source | How each slot is filled | AI? |
|---|---|---|
| AI | Gemini drafts a question for every slot. Requires a Gemini key. | Yes |
| Mix | Alternate slots: even slots from the question bank first, odd slots from Gemini first. | Yes (partly) |
| Custom | The organisation's own question bank first; Gemini only fills slots the bank can't (and only if a key exists). | No - if the bank covers every slot |
| Personal | One assessment per candidate written from the JD + that candidate's resume, using an admin-editable prompt (L1 only). Sections: A 5 MCQs - B 4 experience questions - C 2-3 bug-fixing - D 1-2 DevOps - E 1 Agile. | Yes |

## 2.4 How an AI question is made safe to use

Generation runs in the background: QUEUED -> GENERATING -> VALIDATING -> READY (or FAILED). Nothing Gemini writes is trusted until it passes validation:

- **Code-Write:** the reference solution must pass 100% of tests in every language, and a naive solution must fail at least 2 hidden tests (proves the tests really separate good from weak answers).
- **Code-Debug:** the fixed code passes every test and the buggy code fails at least one hidden test.
- **Code-Output:** the expected output is produced by actually running the snippet.
- **All types:** the question must target a skill in the role profile.
- A failed draft is regenerated with the failure reason - up to **3 Gemini attempts** per slot - then the question bank is tried. A slot nothing can fill is recorded as unfilled, with the reason.
- The previous READY version keeps serving candidates until the new one is READY.
- Bank questions get hourly quality checks; flagged items are excluded from new tests.

**Expert approval:** a job can require a technical reviewer (SME) to approve the question set before any invite can be sent. The recruiter can also edit or regenerate single questions.

# 3. Taking the assessment

- The candidate gets a **single-use, expiring link** (default 48 hours, configurable). The link token is the only credential - no account needed.
- Levels are served in order, one timed attempt each. **Deadlines are enforced by the server**; when time is up, saving and running stop and the attempt is auto-submitted (also by a once-a-minute scheduler).
- "Run" on a coding question executes only the **visible** tests; hidden tests, answers and rubrics never leave the server.
- Code runs in an isolated sandbox (Piston) with no internet, CPU/memory/time limits and a per-question run limit.

## 3.1 Proctoring

Camera and microphone are used in the browser; face and phone detection run locally in the browser (no video frames are sent anywhere). **Proctoring never ends a test** - concerns route the result to HR review instead.

| Warnings (counted by the server) | Review flags (recorded for the reviewer) |
|---|---|
| Tab switch - leaving fullscreen - camera turned off - microphone muted - microphone lost | No face visible - multiple faces - looking away - phone detected - long silence on a spoken question |

# 4. Grading and the final decision

## 4.1 Two grading layers

| Layer | Used for | How |
|---|---|---|
| Objective (no AI) | MCQ, MSQ, Logic, Code-Output; tests of Code-Write and Code-Debug | Exact checks; code is run against ALL tests and scored by the share of hidden tests passed. A compile error or no hidden test passing scores 0. |
| Rubric (AI) | Scenario answers (100%), Code-Write quality (30%), interview answers | Gemini at temperature 0 with a strict schema. Every criterion score must quote evidence from the candidate's own answer - unsupported scores count 0. Graded twice; if the two runs differ by more than 15% of the points, a third run is taken and the median used (flag GRADER_DISAGREEMENT). |

**Anti-gaming rules:** rubric credit is halved when under 50% of hidden tests pass and zero below 25%; hard-coded answers score 0 (flag HARDCODED); comments or the untouched starter code earn nothing.

**When AI is down:** objective scores are kept; AI-graded questions are marked *_PENDING and retried with back-off. After the retry window they become *_UNAVAILABLE and the result goes to human review.

## 4.2 Scores and gates (no AI)

- **Theory** and **Hands-on** = points earned / points possible in that dimension x 100.
- **Overall** = 30% Theory + 70% Hands-on by default (configurable per job).

| Decision | Rule (checked in this order) |
|---|---|
| HR_REVIEW | Any grading uncertainty or integrity concern: grader disagreement, hard-coded code, AI unavailable, or more tab switches than allowed |
| PROCEED | Hands-on >= 60 and Theory >= 50 |
| HR_REVIEW | Hands-on between 50 and 59 |
| NOT_RECOMMENDED | Everything else |

A Theory score 25+ points above Hands-on adds THEORY_PRACTICE_GAP (informational only). A short written summary is produced by Gemini, which may only rephrase the given numbers; if it adds any number not in the input, or AI is unavailable, a fixed template summary is used. A scorecard PDF is available per candidate.

## 4.3 Expert (SME) review

Technical reviewers are assigned per job with a review policy. Results are routed to an SME (or validated by sampling); the SME claims a result, can adjust individual question scores with a reason, and signs off. Signed-off results are locked as VALIDATED, and every action is written to an audit trail. An admin can reopen a review.

# 5. The AI interview

## 5.1 Three interview channels

| Channel | Who runs it | How the candidate answers | Status |
|---|---|---|---|
| Browser interview (BROWSER) | The recruiter starts it from the Candidates page; it runs in the browser in front of them. | Speaks; the browser reads each question aloud and turns the spoken answer into text. | Working |
| Self-service link (SELF) | The recruiter sends a single-use link; the candidate does it alone, any time before it expires. | Same voice experience on the candidate's own device. | Working |
| Phone call (PHONE) | The recruiter clicks Call; Twilio phones the candidate. | Answers by voice on a normal phone call; Twilio speaks the questions and transcribes each answer (with a confidence score). | Built; needs a real Twilio account + public URL, never yet tested with a real call |

## 5.2 Where interview questions come from

- **Recruiter's own questions first** - each job can have its own L1 interview questions, asked in order.
- **Gemini fills the rest** up to the job's total (default **5**, maximum **10**), targeted at the role and the candidate's resume, including **1-2 practical (hands-on) prompts** (2 when Gemini writes 3 or more).
- Every question is one spoken sentence - it is read aloud and answered out loud.
- Gemini also writes the welcome and closing, and 2-4 **expected points** for any recruiter question that has none (kept on the job, editable by HR).
- Interviews can be held in the chosen language (default English - India).

**Without AI: interviews cannot run.** Gemini is called for every interview, even when the recruiter supplies all the questions, because it writes the intro, the outro and the expected points. With no Gemini key the app says: "Add one in Settings before starting an AI interview."

## 5.3 How interviews are graded

- Each answer is graded against its expected points with the same AI rubric as assessments (temperature 0, evidence quoted from the transcript, two runs + median on disagreement): **0, 1 or 2 per point**.
- An empty or very short answer scores 0 without an AI call. Practical prompts count as Hands-on.
- The result uses the same Theory/Hands-on scores and gates as assessments, then goes to SME review.
- If Gemini is unreachable the answer is retried, then marked unavailable - which forces HR review.

# 6. With AI vs without AI - summary

| Capability | With Gemini | Without Gemini |
|---|---|---|
| Role profile | Read from the JD by AI | Rule-based from extracted skills + experience |
| Resume screening | Same - rule-based | Works fully (never uses AI) |
| Recruiter chat assistant | Understands requests, runs screening | Not available |
| Assessment - AI source | Every slot written + validated | Refused (needs a key) |
| Assessment - Mix source | Bank + AI alternate | Bank only, if it can cover the slots |
| Assessment - Custom source | Bank first, AI fills gaps | Works fully from the bank |
| Personal assessment | Written per candidate | Not available |
| MCQ / MSQ / Logic / Code-Output grading | Rules | Works fully |
| Code-Write / Code-Debug tests | Sandbox | Works fully (needs the Piston runner) |
| Code-Write quality (30%) + Scenario answers | AI rubric | Pending, then human review |
| Result summary text | AI rephrasing | Fixed template |
| Decision gates + scorecard | Rules | Works fully |
| Interviews (all 3 channels) | Full | Not available |
| Interview grading | AI rubric | Pending, then human review |
| Proctoring | Browser-side, rules | Works fully (never uses AI) |
| Expert (SME) review | Human | Works fully |

## 6.1 In one sentence

Viztalent AI can screen resumes, run proctored tests from its own question bank, auto-grade objective and coding questions, and produce decisions **without any AI**; Gemini adds automatic question writing, personal tests, AI interviews, rubric grading of written/spoken answers and summaries - and wherever AI is used, its output is validated, evidence-checked, and backed by a human-review fallback.

## 6.2 Notes and limits

- Phone interviews are fully built but have never been run against a real Twilio call.
- Interview video recording is designed (INTERVIEW_CHANNELS_DESIGN.md) but not built; interviews keep transcripts.
- Coding questions need the Piston code-runner service with languages installed.
- AI usage is capped per day (AI_DAILY_CALL_CAP, default 2000 calls) and can fall back to a second Gemini model.